

Alex J. Mercer

Copenhagen, Denmark

 <https://github.com/alexmercer-ai>

 <https://linkedin.com/in/alexjmercer>

+45 20 12 34 56

alex.mercer@email.com

Summary

Forward-Deployed AI Engineer with over 6 years of experience translating cutting-edge artificial intelligence into production-grade solutions that drive tangible business and healthcare impact. Proven expertise in architecting and deploying LLM-based applications, semantic search pipelines, and autonomous agent frameworks from ideation to scalable production. Adept at collaborating across multidisciplinary teams of data scientists, domain experts, and platform engineers to re-imagine enterprise processes using modern AI standards (MCP, A2A, OpenTelemetry) and cloud-native infrastructures (AWS, Azure, Databricks, Snowflake). Combines strong software engineering fundamentals with continuous learning in generative AI, CI/CD governance, and responsible AI practices. Deeply motivated by leveraging data and AI to solve complex chronic disease challenges, delivering high-quality, patient-centric solutions in dynamic, multinational environments.

Work Experience

- PharmaTech Innovations** Copenhagen, Denmark
Senior AI Engineer 2021-03 - Now
 - Spearheaded the end-to-end development of an AI-powered clinical trial matching platform, reducing patient enrollment time by 40% through semantic search and LLM-driven document parsing.
 - Designed and implemented a modular AI Accelerator framework that standardized the translation of research prototypes into production-ready microservices across 3 business units.
 - Collaborated with data scientists and domain experts to define evaluation benchmarks for AI agents, improving model reliability and safety metrics by 28%.
 - Integrated modern AI communication standards (MCP, A2A) and observability tools (OpenTelemetry) into the enterprise stack, enabling seamless cross-system agent interactions.
 - Established CI/CD pipelines using GitHub Actions and Terraform, automating testing, containerization, and deployment of AI workloads on AWS and Azure.
 - Mentored a team of 5 junior engineers in best practices for agentic IDE integrations (Cursor, VS Code), rapidly prototyping tools and reducing development cycles by 30%.
- DataScale Solutions** Copenhagen, Denmark
Machine Learning Engineer 2018-06 - 2021-02
 - Architected and deployed vector database infrastructures (Pinecone, Weaviate) to support enterprise semantic search and RAG pipelines processing millions of daily queries.
 - Co-built reusable infrastructure components and evaluation frameworks in partnership with Platform AI teams, enabling company-wide rollout of generative AI capabilities.
 - Optimized LLM inference pipelines through quantization and prompt engineering, reducing latency by 55% and infrastructure costs by 22%.
 - Partnered with cross-functional stakeholders to identify automation opportunities, successfully migrating 12 legacy business processes to AI-driven workflows.

- Maintained rigorous code quality and documentation standards, fostering a culture of continuous feedback and agile iteration in a remote-first, multinational team.

- **Nordic Cloud Services** Copenhagen, Denmark
Software Engineer (AI/ML Track) 2016-09 - 2018-05
 - Developed backend microservices and data pipelines supporting early-stage machine learning models for predictive analytics and customer segmentation.
 - Implemented automated testing and monitoring frameworks that improved system uptime to 99.9% and streamlined deployment workflows.
 - Actively participated in hackathons and innovation sprints, delivering rapid prototypes that were later integrated into the company's core product line.
 - Gained foundational experience in cloud computing, containerization (Docker, Kubernetes), and version control, laying the groundwork for advanced AI engineering.

Projects

- **Enterprise AI Agent Framework & Evaluation Benchmarks** 2022-09 - 2023-08
Lead AI Engineer
 - Led the design and implementation of a multi-agent orchestration platform using MCP and A2A standards, enabling secure, standardized communication between internal AI services.
 - Developed comprehensive evaluation benchmarks for agent performance, safety, and alignment, directly feeding into product AI engineering roadmaps.
 - Collaborated with platform engineers to containerize and deploy the framework on AWS ECS, supporting over 50 internal teams.
 - Produced detailed technical documentation and training materials to drive internal adoption and upskilling across engineering departments.
- **Healthcare Document Intelligence Pipeline** 2020-04 - 2021-05
Senior AI Engineer
 - Built a RAG-based document processing system leveraging Snowflake for data orchestration and Azure AI Search for semantic retrieval.
 - Integrated modern IDE integrations and agentic coding assistants to accelerate prototyping of NLP components for medical record parsing.
 - Reduced manual document review time by 65% while maintaining 98.5% accuracy against expert-labeled ground truth datasets.
 - Ensured compliance with data governance and responsible AI principles by implementing audit trails, data masking, and model monitoring dashboards.

Education

- **Technical University of Denmark (DTU)** Copenhagen, Denmark
MSc Data Science & Artificial Intelligence 2014-08 - 2016-06
- **Aalborg University** Copenhagen, Denmark
BSc Computer Engineering 2010-08 - 2014-07

Courses & Certifications

- **Advanced Large Language Models & Prompt Engineering** DeepLearning.AI
2023-05
- **Cloud-Native AI Platform Architecture** AWS & Databricks Academy
2022-11
- **Responsible AI & AI Governance Frameworks** Stanford Online
2022-03
- **Agentic AI Development & MCP Standards** Coursera (University of Michigan)
2023-09
- **CI/CD & DevOps for ML Workloads** Kubeflow & GitHub
2021-08

Technical Skills

Generative AI & LLMs, AI Agent Frameworks & Orchestration, MCP (Model Context Protocol), A2A (Agent-to-Agent) Standards, Vector Databases & Semantic Search, RAG Architectures, OpenTelemetry & AI Observability, Cloud-Native AI (AWS, Azure, GCP), Databricks & Snowflake, CI/CD for ML (GitHub Actions, MLflow, DVC), Containerization & Orchestration (Docker, Kubernetes), Python, Rust, Go, Agentic IDE Integrations (Cursor, VS Code, Copilot, Codex), Responsible AI & Governance, Model Evaluation & Benchmarking, Prompt Engineering & Fine-tuning, RESTful APIs & Microservices, Data Pipeline Engineering, Agile & Scrum Methodologies, Technical Stakeholder Management, Cross-Functional Team Leadership, Rapid Prototyping & Innovation, Documentation & Technical Writing

Languages

English: Fluent (C2)
Danish: Professional Working Proficiency (B2)
German: Conversational (B1)

Patents

- **Method and System for Dynamic Agent Communication in Distributed AI Environments** DK102345A1
Granted 2023-02
- **Automated Benchmark Generation Framework for LLM-Based Applications** US2024089123A1
Under Examination 2024-01

Publications

- **Scalable Evaluation Frameworks for Enterprise AI Agents: Metrics, Benchmarks, and Deployment** 2023-10
- **Bridging Research and Production: A Practical Guide to AI Accelerator Frameworks in Healthcare** 2022-06

- Optimizing Inference Costs for LLM-Based RAG Systems in
- Cloud Environments

2021-09